Topic 44 — Current Environmental Issues

Covers syllabus (click to expand)

Carbon Footprint | Ecological Footprint (पारिस्थितिक पदचिह्न) | Net Zero Emissions (नेट जीरो उत्सर्जन) | Green Economy (हरित अर्थव्यवस्था) | Blue Economy (नीली अर्थव्यवस्था) | Circular Economy (चक्रीय अर्थव्यवस्था) | Green Hydrogen | Microplastics | Climate Refugees | Nature-based Solutions | LiFE Mission

Sources baked in: IPCC; UNEP; Global Footprint Network; MoEFCC; MNRE; IUCN NbS; UPPCS PYQs 2018–2025 **Weight:** ★★★ High — LiFE COP trap; MISHTI year; India net zero 2070; microplastic <5 mm; EF (gha) ≠ carbon (tCO_2e) **Last verified:** August 2026

Current Affairs (this topic)

- Primary living CA chapter — NbS, plastics, heat, Himalaya (हिमालय) cryosphere, LiFE.
- Cross-link facts to Climate / Biodiversity / Pollution without relocating treaty years.

44.1 Carbon Footprint

Identity: Total GHGs as tCO_2e | Scope 1/2/3 | India low per capita, high total

- A **carbon footprint** is the total greenhouse gases caused directly and indirectly by a person, organisation, product, or nation, expressed as **tonnes of CO_2 equivalent (tCO_2e)**.
- **Scope 1** covers **direct** emissions from owned sources (boilers, fleet fuel).
- **Scope 2** covers **purchased electricity, heat, or steam**.
- **Scope 3** covers other indirect emissions (supply chain, product use, travel, waste) — often the largest share and hardest to measure.
- India's **per capita** footprint is about **~2 tCO_2e**, below the global average (~4.7) and far below the USA (~14+), even though India's **absolute** emissions rank near the top globally.
- Energy/electricity is the largest Indian sector share; diet, transport mode, and grid mix drive lifestyle footprints.
- Carbon footprint is **not** the same as ecological footprint (land-area metric in **gha**).

Scope	Type	Example
Scope 1	Direct	Factory (फैक्टरी) boiler
Scope 2	Purchased energy	Grid electricity
Scope 3	Other indirect	Supplier emissions, flights

> **Logic:** "India has the highest per capita carbon footprint" = **FALSE**. Unit = **tCO_2e**, not gha.

44.2 Ecological Footprint

Identity: gha unit | Biocapacity | Global Footprint Network | Broader than carbon

- An **ecological footprint** measures how much biologically productive land and water a population (जनसंख्या) needs for resources and waste absorption, in **global hectares (gha)**.
- **Biocapacity** is nature's regenerative capacity; if footprint exceeds biocapacity, there is an **ecological deficit**.
- **Earth Overshoot Day** marks when annual demand exceeds that year's regeneration.
- Components include cropland, grazing land, forest, fishing grounds, built-up land, and carbon-absorption land.
- India per capita ecological footprint is roughly **~1.0–1.2 gha** — below the global average (~2.8 gha) but rising with urban consumption.
- Standard methodology is associated with the **Global Footprint Network**.
- shorthand: ecological footprint measures **human pressure on nature** — broader than GHG-only carbon accounting.

Metric	Unit	Rough India
Ecological footprint	gha	~1.0–1.2 / person
Carbon footprint	tCO_2e	~2 / person

> **Logic:** Same unit for both footprints = **FALSE**. EF = **gha**; carbon = **tCO_2e**.

Social Cost of Carbon

- The **Social Cost of Carbon (SCC)** is a **monetary measure** of the long-term damage done by emitting **one tonne of CO_2** in a given year.
- US EPA and other agencies use SCC estimates (प्राकलन समिति) to evaluate climate impacts of rule-making.
- SCC is **not** the same as a person's carbon footprint, fossil-fuel requirement of a country, or climate-refugee adaptation cost.

44.3 Net Zero Emissions

Identity: Emissions = removals | India 2070 COP26 | NDC 2030 | Gap Report = UNEP

- **Net zero** means anthropogenic emissions are **balanced by anthropogenic removals** over a period — residual emissions may remain if matched by sinks or removals.

- **Gross zero** means eliminating virtually all emissions — harder than net zero.
- India pledged **net zero by 2070** at **COP26 Glasgow (November 2021)**.
- Updated **NDC 2030:45%** emissions-intensity cut vs 2005; **50%** non-fossil electricity capacity; additional **2.5–3 $GtCO_2e$** forest/tree-cover sink.
- Country ladder often tested: EU/UK **2050**, China **2060**, India **2070**.
- **UNEP Emissions Gap Report** measures the gap between pledges and 1.5°C/2°C pathways.

Country/bloc	Net zero year
EU / UK	2050
China	2060
India	**2070**

PYQ — UPPCS Prelims 2021, Q70

With reference to 'NET-ZERO' goal for 2050: (1) It means emissions brought down to zero by 2050. (2) Emissions will be compensated by absorption and removal of GHGs.

Show answer

Ans: B (Only 2) — Stmt 1 describes **gross zero**; net zero allows balance via removals.

PYQ — UPPCS Prelims 2024, Q6

Match: Emissions Gap Report → **UNEP**.

Show answer

Ans: A (3 1 4 2) — HDR=UNDP (यूएनडीपी); WEO=IMF; WIR=UNCTAD; Emissions Gap=**UNEP**.

> **Logic:** India net zero = **2070**, not 2050. Gap Report ≠ IPCC (आईपीसीसी) Assessment Report.

44.4 Green Economy

Identity: UNEP 2011 | Low-carbon + inclusive | Decouple GDP from harm | Broader than circular

- UNEP's **green economy** (2011 framing) improves well-being and equity while cutting environmental risks — **low-carbon, resource-efficient, socially inclusive**.
- Goal is to **decouple GDP growth from environmental degradation (पर्यावरणीय क्षरण)**, not to stop growth.
- India levers include renewables (**500 GW non-fossil capacity by 2030** pledge track), EVs (FAME-II), efficiency (PAT/UJALA), and green hydrogen.
- **Green finance** (हरित वित्त) (sovereign green bonds, SEBI green-bond rules) funds this pathway — detailed market tools sit with Topic 24.

- Green economy is **broader** than circular economy; circular material loops are one component.

> **Logic:** Green economy ≠ circular economy as identical labels.

44.5 Blue Economy

Identity: SDG 14 | Ocean livelihoods + conservation | MISHTI 2023 | Sagarmala | Blue carbon (नीला कार्बन)

- **Blue economy** is sustainable use of ocean and coastal resources for growth and jobs while protecting marine ecosystems.
- **SDG 14** = **Life Below Water**.
- India has a large **EEZ (~2.37 million sq km)** — fisheries, ports, shipping, coastal tourism, offshore wind, marine biotech.
- **Blue carbon** is carbon stored in mangroves, seagrass, and salt (लवणाध्यक्ष) marshes.
- **MISHTI** (Mangrove Initiative for Shoreline Habitats and Tangible Income) was announced in **Union Budget 2023–24** to restore mangroves for shoreline protection and income.
- **Sagarmala** is port-led coastal development; **Deep Ocean Mission** covers deep-sea exploration (Samudrayaan).
- UP is **landlocked** — blue-economy PYQs are national schemes, not UP coastal projects.

PYQ — UPPCS Prelims 2025, Q93

MISHTI: (1) Initiated in 2019. (2) Aims to restore mangrove forests.

Show answer

Ans: A (Only 2) — Restoration true; year is **2023–24 Budget**, not 2019.

Sector	Fact
Mangroves	MISHTI 2023
Ports	Sagarmala
Deep sea	Deep Ocean Mission

> **Logic:** MISHTI = coastal mangroves, **not** Ganga-basin inland planting.

44.6 Circular Economy

Identity: Close loops | 3R/9R | EPR rules | SDG 12 | ⊂ green economy

- A **circular economy** designs out waste by keeping materials in use through reduce, reuse, repair, remanufacture, and recycle — opposite of linear **take-make-dispose**.

- Classic **3R** expands in some frameworks to a fuller **9R** ladder (refuse → recover).
- India operationalises circularity mainly through **Extended Producer Responsibility (EPR)** (विस्तारित उत्पादक दायित्व) — plastic, e-waste, batteries, tyres.
- **Plastic Waste Management Rules 2016** (amended 2022) and related EPR rules make producers responsible for end-of-life collection.
- Circular economy maps strongly to **SDG 12** (Responsible Consumption and Production).
- It is a **component** of green economy, not a synonym for the whole green-growth agenda.

Linear	Circular
Virgin in → dump out	Recycled/regenerative loops
Waste = disposal problem	Waste = design flaw

> **Logic:** "Circular = green economy identically" = **FALSE**.

44.7 Green Hydrogen

Identity: Renewable electrolysis | Mission Jan 2023 | 5 MMT by 2030 | Colour codes

- **Green hydrogen** is H_2 from **water electrolysis** powered by **renewable electricity**.
- Hydrogen is an **energy carrier**, not a primary source; at point of use it burns to water vapour.
- **Grey hydrogen** from natural-gas reforming dominates today's supply and is **not** climate-friendly.
- **Blue hydrogen** is grey production plus **CCS** — a debated transition label.
- **National Green Hydrogen Mission (January 2023)**, coordinated under **MNRE**, targets about **5 MMT/year by 2030** for refining, ammonia/fertiliser, steel, and heavy transport.
- Grey ≠ green — the colour word is an trap.

Colour	Method
Green	Electrolysis + renewables
Grey	Natural gas reforming
Blue	Grey + CCS
Black/brown	Coal gasification

> **Logic:** Mission = **2023**, not 2019. Target = **5 MMT by 2030**.

44.8 Microplastics

Identity: <5 mm | Primary vs secondary | Persistence | SUP ban 2022 | Ganga (गंगा) pathway

- **Microplastics** are plastic particles **smaller than 5 mm**.
- **Primary** types are manufactured small (microbeads, nurdles).
- **Secondary** types form when larger plastics fragment (bags, nets, tyre wear) — most environmental mass is secondary.
- Pathways run land → rivers → sea; Ganga/Yamuna (यमुना) studies show urban downstream contamination.
- Plastics photodegrade into **smaller** particles; they do **not** vanish into harmless minerals in weeks.
- India banned identified **single-use plastic items in July 2022** and uses EPR under plastic-waste rules.
- UNEA **2022** launched Global Plastics Treaty negotiations (INC process).

Type	Origin
Primary	Made small
Secondary	Fragmented from large items

> **Logic:** Threshold = **<5 mm**, not 5 cm. “Biodegrade in weeks” = **FALSE**.

44.9 Climate Refugees

Identity: Climate-linked mobility | Mostly internal | No 1951 legal category | Bundelkhand UP angle

- **Climate refugees / climate migrants** are people forced to move by climate impacts — cyclones, floods, drought (सूखा), sea-level rise, desertification (मरुस्थलीकरण).
- Preferred policy language often uses **climate migration**; **“climate refugee”** is **not** a formal category under the **1951 Refugee Convention** (persecution-based).
- Most climate-linked displacement is **internal** within countries, not cross-border asylum claims.
- Sudden-onset (floods, cyclones) differs from slow-onset (drought, sea-level rise).
- UP example: **Bundelkhand** drought stress drives outmigration toward cities.
- Mobility is **multi-causal** — climate interacts with poverty and livelihoods; not every migrant is “climate-only.”

Type	Example
Sudden-onset	Cyclone / flash flood displacement
Slow-onset	Drought / sea-level rise
Legal refugee (1951)	Persecution — not climate alone

> **Logic:** No dedicated Indian **Climate Refugee Act**. Most Indian cases = **IDPs / internal migrants**.

44.10 Nature-based Solutions (NbS) (प्रकृति-आधारित समाधान)

Identity: IUCN | Protect–manage–restore | Climate + society | MISHTI | ≠ geoengineering

- **Nature-based Solutions** protect, sustainably manage, or restore ecosystems to address societal challenges (climate, disasters, food, water) while aiding biodiversity and well-being.
- Mitigation NbS: forests, mangroves (blue carbon), soils, peatlands.
- Adaptation NbS: mangrove storm buffers, wetlands for flood peaks, urban greenery for heat.
- **MISHTI mangrove restoration** is a flagship Indian NbS example (Budget **2023–24**).
- Related India tracks: Green India Mission, Amrit Dharohar wetlands, Namami Gange (नमामि गंगे) riparian works.
- NbS ≠ **geoengineering**; planting trees is ecosystem management, not solar-radiation engineering.
- Geoengineering teaching contrast: **cirrus cloud thinning** and **stratospheric sulphate aerosol injection** are proposed to cool the planet / offset warming — they are **not** NbS and are not for cyclone control or solar-wind shielding.
- NbS may generate carbon credits but the concept is broader than offset markets alone.

> **Logic:** NbS = ecosystems for people + planet. Permanence risk (fire, clearing) still matters for carbon claims.

44.11 LiFE Mission

Identity: Lifestyle for Environment (जीवनशैली पर्यावरण) | MoEFCC | COP26 idea 2021 | Launch June 2022 | 75 actions

- **LiFE (Lifestyle for Environment)** nudges individual and community behaviour toward mindful consumption.
- The idea was proposed at **COP26 Glasgow, November 2021**.
- **Mission LiFE** was formally launched in **June 2022**; **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) is nodal.
- Catalogue of about **75** pro-planet actions covers energy, water, waste, food, and e-waste habits.
- It is a **behavioural nudge** programme — not a substitute for solar missions, carbon markets, or Green Credit rules.
- Supports demand-side cuts that help the **net zero 2070** pathway.

PYQ — UPPCS Prelims 2025, Q139

LiFE: (1) Launched in June 2022. (2) Idea proposed at COP-25 in Madrid.

Show answer

Ans: D (Only 1) — Launch true; idea = **COP26 Glasgow 2021**, not COP-25 Madrid 2019.

Field	Detail
Idea	COP26 **2021**
Launch	**June 2022**
Nodal	**MoEFCC**
Actions	**~75**

> **Logic:** COP-25 Madrid trap is the classic false statement.
